



Exploring Accent Variability, Phrase Difficulty, and Speaker Distance in Robot-Enabled Speech Recognition

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Introduction

- **Audio Speech Recognition (ASR)** is a growing field that helps hearing-impaired individuals understand conversation around them
- **Existing ASR Models:** Amazon Transcribe, Apple, Google, IBM Watson Speech to Text, etc.
- **Our Goal:** Investigate what happens when one of the available open-source models is deployed on a robot and determine how well it performs on:
 - Various accents
 - Phrase difficulties
 - Speaker distances

Relevant Work

“Evaluation of Off-the-shelf Speech Recognizers on Different Accents in a Dialogue Domain” (Tadimeti et al., 2022)	“Social Robotics Meets Sociolinguistics: Investigating Accent Bias and Social Context in HRI” (Foster et al., 2023)
• Analyzes how well ASR models/APIs perform Amazon, IBM, Google, Microsoft, and Apple • Found a performance drop for accented English (i.e., British, Indian, French, East Asian, etc.)	• Researchers found that the robot had difficulty with translating English spoken with a Scottish accent. • Younger, middle-class, male participants were more positive overall about the interaction than other groups.

Research Gap

Much of the existing literature on accented Audio Speech Recognition has one or more of the following limitations:

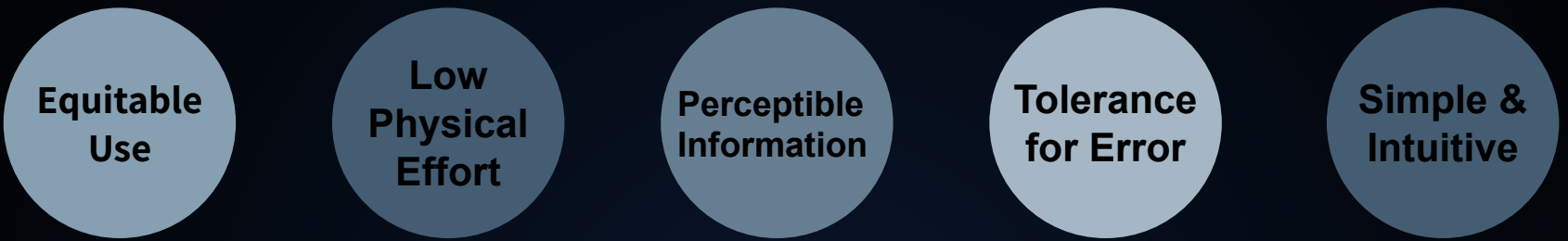
Objectively analyzes model performance but not interactions between human and software/robot

ASR model is not deployed on an embodied robot

ASR model is only tested on a single accent group

ASR performance not analyzed from the perspective of phrase difficulty and speaker volume/distance

Our work addresses all of these limitations and further analyzes the use of an ASR assistive robot from the lens of educational settings, such as for a hearing-impaired speaker in a classroom lecture.



**Equitable
Use**

**Low
Physical
Effort**

**Perceptible
Information**

**Tolerance
for Error**

**Simple &
Intuitive**

Our View = Social Model + Medical Model

People are disabled by both medical conditions and social barriers. Hearing-impaired individuals currently lack access to a robust assistive device that can effectively help them interpret conversations around them.

Experiment Overview

Independent Variables

- **Accents:** American (2 rounds), Chinese (1 round), Guyanese (1 round), and Brazilian (1 round) accents were studied
- **Listener Types:** Hearing-impaired humans, non hearing-impaired individuals, Pepper robot
- **Speech Complexity:** Two “easy” phrases and two “hard” phrases per speaker
- **Speaker Distance:** One “close” speaker and one “far” speaker per round

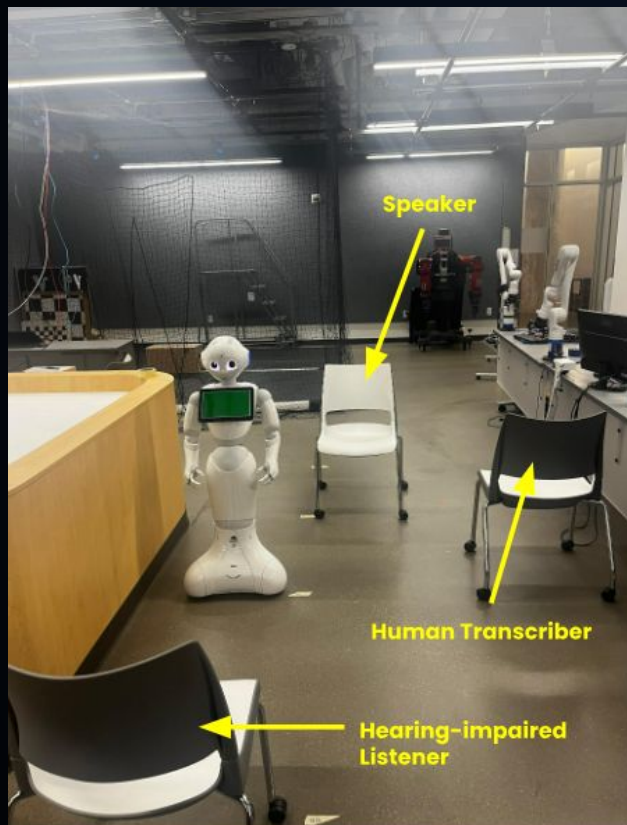
Dependent Variables

- **Translation Accuracy:** Measured by the Translation Error Rate (TER) of the human vs. robot.
- **Listener Comprehension:** The level of comprehension by the hearing-impaired listener.

Experiment Overview

- **Hypothesis 1:** The translation accuracy and listener comprehension levels will be lower for non-American accents compared to American accents.
- **Hypothesis 2:** ASR performance and listener comprehension will decrease with increased speaker distance and phrase difficulty.

Methods: Procedure & Experimental Setup



- **Chosen Model: Google Speech Recognition**
- Open-source, available in GitHub, easiest to integrate with Pepper's NaoQi SDK
- Study divided into 5 rounds, each of which focused on a certain non-American accent
- Each round consisted of a human interpreter, Pepper as a robotic interpreter, a “far” speaker, “close” speaker, and a hearing-impaired listener
- Each speaker spoke 2 “easy” phrases and 2 “difficult” phrases to the robot



The robot makes more translation errors than humans

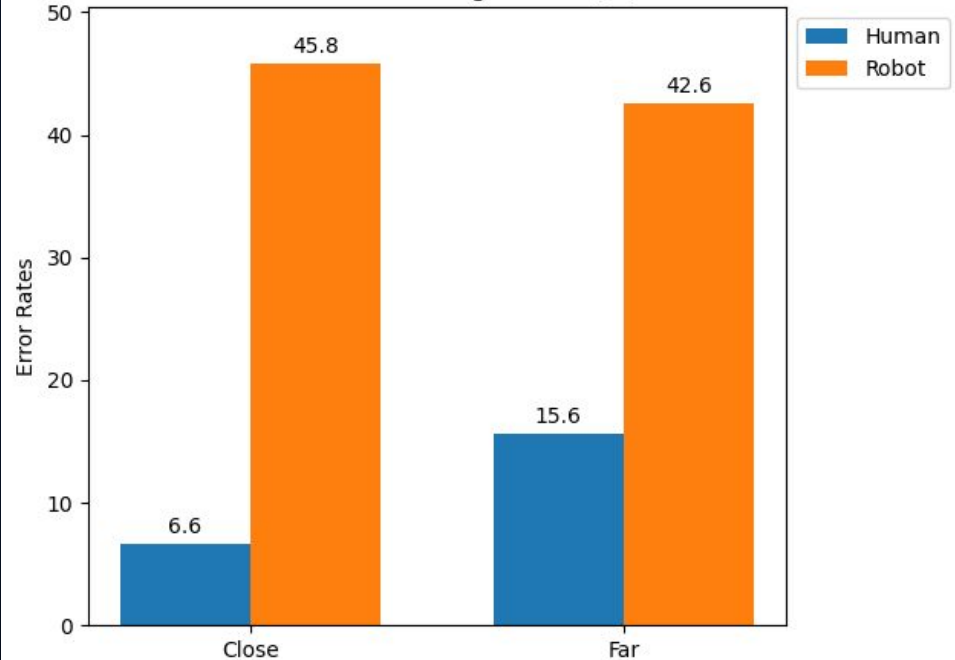
Average Errors for Close Distance (%)

	Average Human Error	Average Robot Error	Average Error Difference
Round 1	2.00	22.00	20.00
Round 2	4.00	51.00	48.00
Round 3	18.00	49.00	31.00
Round 4	1.00	83.00	82.00
Round 5	8.00	24.00	16.00

Average Errors for Far Distance (%)

	Average Human Error	Average Robot Error	Average Error Difference
Round 1	7.00	54.00	47.00
Round 2	23.00	38.00	24.00
Round 3	2.00	47.00	45.00
Round 4	4.00	51.00	47.00
Round 5	42.00	23.00	44.00

Bar Chart of Average Errors (%)

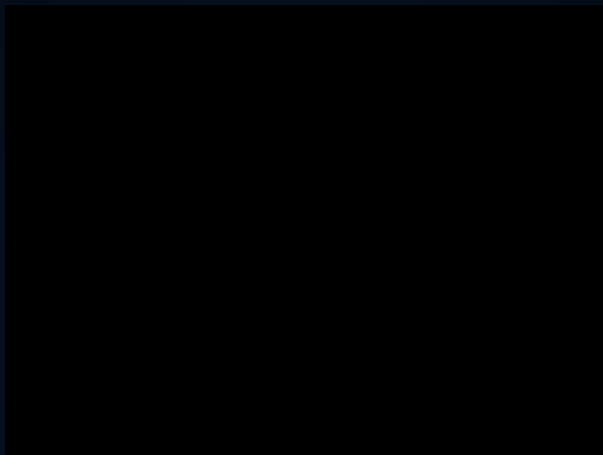


American Accent



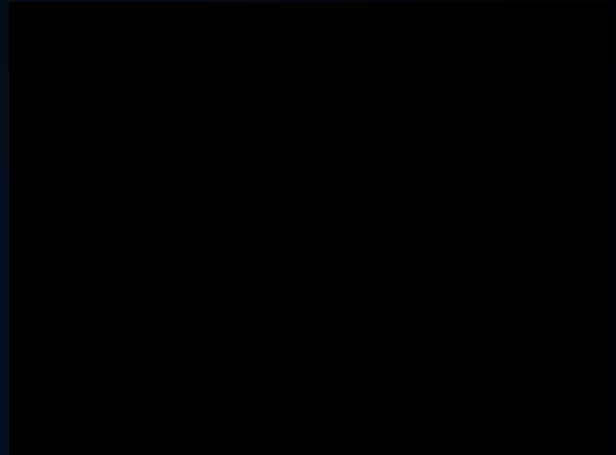
Close

Chinese Accent



Far

Brazilian Accent



Far



“

“As a listener I was hoping to see the **translation real time**. I was also noticing that sometimes I was naturally trying to read the lips of the person talking instead looking at the robot because I **wanted to see the person’s facial expression.**”

“I think it could do very well and be extremely helpful if it has a little bit more time to train and improve its model... As a speaker I was **making sure I was loud and speaking a little more slowly** than I usually would.”

Insights

Users reported...

- Looking back and forth between the display screen and the speaker's face to analyze their emotions
- Having to repeat phrases because they weren't loud enough for the robot to hear was frustrating
- Small errors in the transcription caused them to not be able to make sense of the sentence spoken
- Feeling like they had to talk more slowly and loudly for the robot to understand

Conclusions

1. The translation accuracy and listener comprehension levels were lower for non-American accents compared to American accents.
2. ASR performance and listener comprehension decreased with increased speaker distance and phrase difficulty.



Challenges & Limitations

- Initially planned to perform study on impact of accents on Visual Speech Recognition models (lipreading), but incompatible with Pepper's NaoQi OS
- Two of the accents (Guyanese and Chinese) were impersonated and not real
- Each speaker was assumed to represent an entire accent group, not accounting for variations in accents within the accent group
- Participants were not actually hearing-impaired, wore noise-canceling headphones
- Limited number of participants
- Actual difficulty of “easy” and “hard” phrases is subjective

Future Work

1

Perform the study on a larger scale: more accent groups, more participants per accent group

2

Develop more advanced ASR models with improvements such as: increased signal-to-noise ratio, reduced translation delay, and incorporation of punctuation & semantics

3

Do a performance comparison between various ASR models

4

Analyze whether hearing-impaired users prefer embodied robots that can perform VSR or ASR



THANKS!

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References

Papers

- Foster, M. E. and Stuart-Smith, J. (2023) Social Robotics Meets Sociolinguistics: Investigating Accent Bias and Social Context in HRI. In: 18th Annual ACM/IEEE International Conference on Human Robot Interaction (HRI2023), Stockholm, Sweden, 13-16 March 2023, pp. 156-160. ISBN 9781450399708
- Tadimeti D., Georgila K., Traum D. (2022) Evaluation of Off-the-shelf Speech Recognizers on Different Accents in a Dialogue Domain. Proceedings of the 13th Conference on Language Resources and Evaluation (LREC 2022), pages 6001–6008 Marseille, 20-25 June 2022.

Links

- <https://github.com/JBramauer/pepperspeechrecognition>

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